



Occlusion-Robust Face Identification with Partially Fine-Tuned ArcFace

Machine Learning Laboratory – CSE 4112

Department of Computer Science & Engineering

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Research Question

Does occlusion robustness transfer to identities never seen during fine-tuning?

The task

Identify an enrolled person from an occluded face.

Compare three adaptation scopes using the same gallery and decision threshold.

A correct label below the threshold is still a rejection.

The experiment

E0 · Frozen pretrained ArcFace

E1 · Fine-tune the final residual block

E2 · Fine-tune the block + embedding head

20 train / 5 validation / 5 test identities per fold.

Person-disjoint fine-tuning evaluation; all 30 identities remain enrolled in the gallery.

Dataset & Occlusion Conditions

In-person capture, signed consent, pseudonymous identities p01–p30.

30

Identities

6 collectors × 5 people

6,000

Raw images

200 per person

5,988

Aligned crops

99.8% detection

8 × 2

Conditions

occlusion × lighting

Capture protocol

Phone-camera collection in regular and low light. Single-sitting capture; indoor conditions.

Evaluation pool

5,818 queries after 170 gallery / near-burst exclusions. Fixed gallery: 30 identities.

[Dataset folder](#) · [consent and access restrictions still apply](#)

Eight Occlusion Conditions

Representative participant images reused from the previous presentation.



None



Cap



Clear glasses



Sunglasses



Scarf



Mask



Mask + clear glasses



Mask + sunglasses

Why This Dataset?

Pgu-Face is the base dataset paper—not a directly comparable recognition benchmark.

Property	Pgu-Face (2016)	Our dataset
Participants	224	30
Images / person	4 (896 total)	200 raw (6,000 total)
Covering variation	Facial hair + sunglasses	8 types, including compound masks
Capture conditions	Up to 2 sessions, ≥ 6 days apart	Regular + low light; one sitting
Evaluation artifacts	Partially covered-face collection	112 $\times$ 112 crops, labels, six-fold split

More within-person variation and coverings; fewer identities and no across-session evaluation.

Different—and Better for This Question

Broader occlusion stress testing, rather than a claim of universally better data.

What the new data enables

~6.7× as many raw images overall.

50× as many captures per identity.

Mask + glasses combinations probe severe, compound occlusion.

Two light levels and condition labels support targeted error analysis.

What Pgu-Face still offers

224 identities versus our 30: greater identity diversity.

Separated capture sessions probe time-related change.

Different data and protocols mean published accuracies cannot be ranked directly.

The advantage is alignment with our occlusion-transfer study—not broader population coverage.

Curation, Quality & Provenance

Retain the documented dataset; disclose limitations rather than inventing cleanup counts.

Processing checks

12 raw images failed detection.

5,988 crops: 3,050 regular / 2,938 low light.

429 duplicate MD5 occurrences retained, all within the same identity.

Person-disjoint folds keep a person's copies together.

Coverage & consent

Scarf coverage: 24 of 30 people; some other condition gaps.

A small, unflagged digitally added covering subset was reported previously.

Academic, non-commercial, withdrawable consent; faces are still identifiable.

No claim of perfectly balanced coverage, fully natural coverings, or an unrestricted public license.

Detection Failures: Actual Examples

Five documented rejected raw images: examples 1, 2, 5, 6 and 7 from the report.

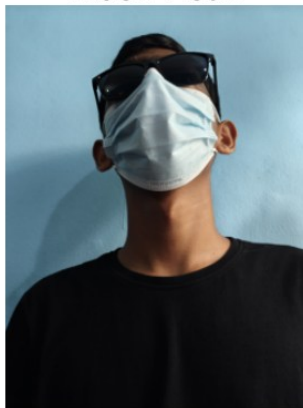
1: p03
Mask + sun



2: p07
Mask + sun



5: p14
Mask + sun



6: p14
Mask + sun



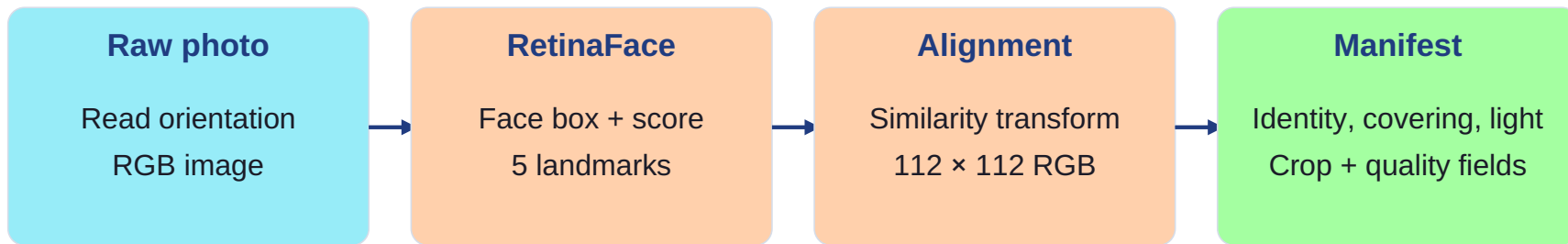
7: p16
Mask



RetinaFace detects 5,988 of 6,000 raw images. These failures never enter the recognition evaluation.

Preprocessing Pipeline

Frozen RetinaFace detection and landmark alignment produce the recognizer's inputs.

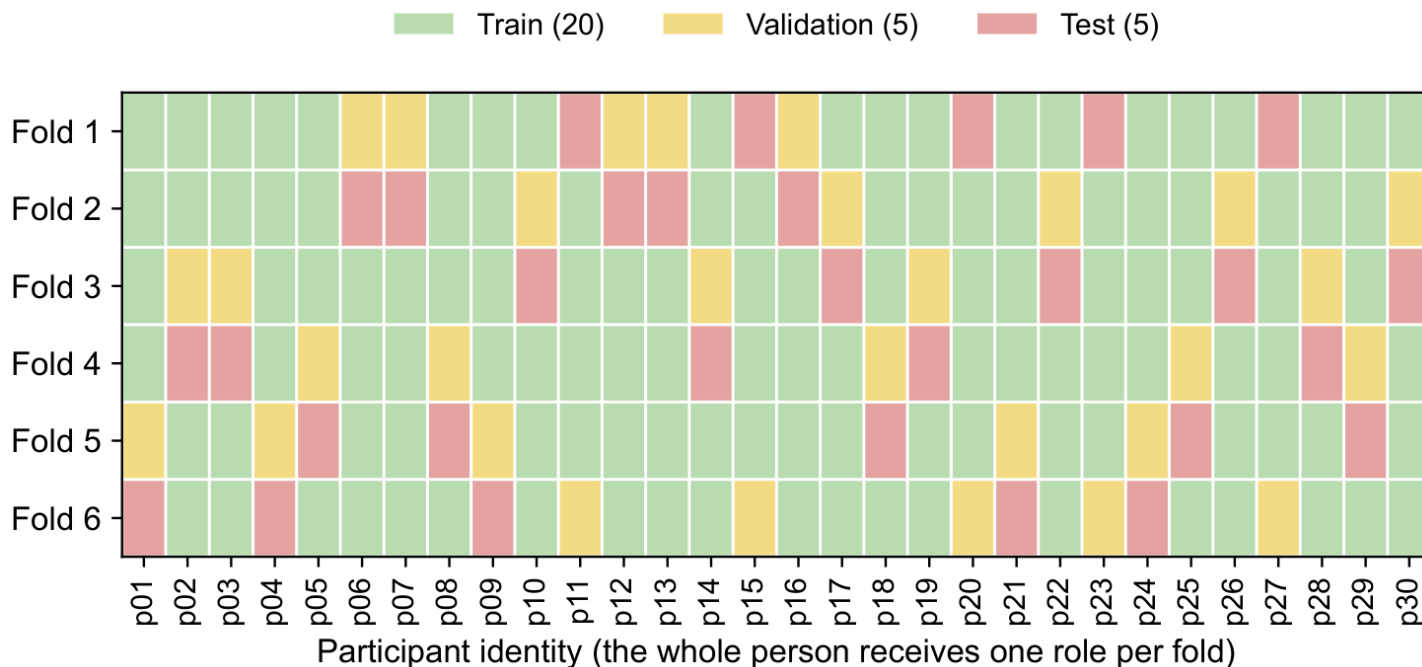


Recognition normalization

$(\text{pixel} - 127.5) / 127.5$; channel-first tensor $N \times 3 \times 112 \times 112$.

Person-Disjoint Six-Fold Evaluation

Seed 42 · 20 train identities, 5 validation identities, 5 held-out test identities per fold.



Every identity is tested once. Validation uses the next five-person block cyclically.

One Fixed Gallery, Fair Comparisons

Enrollment is not fine-tuning: held-out test identities have a gallery reference only.

Build once with E0

Choose the highest-detection-score “none” crop per identity.

Compute and L2-normalize its pretrained 512-D embedding.

Lock these 30 reference vectors across E0, E1, and E2.

Prevent shortcut evaluation

Exclude gallery photos and near-burst queries (170 images total).

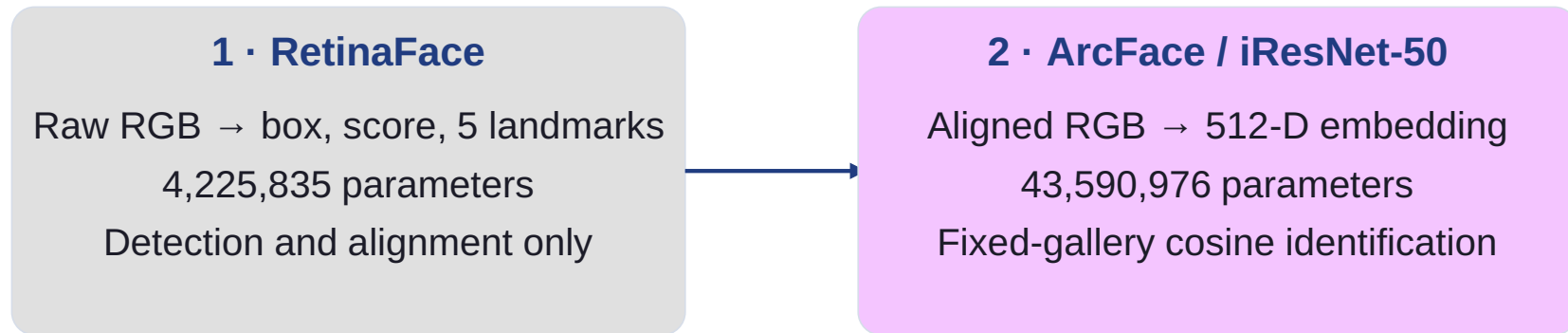
Fine-tuning uses only the 20 training identities' fixed vectors.

Validation and testing match against all 30 gallery identities.

5,818 pooled test queries. No held-out person contributes fine-tuning gradients.

Two Models, Two Distinct Roles

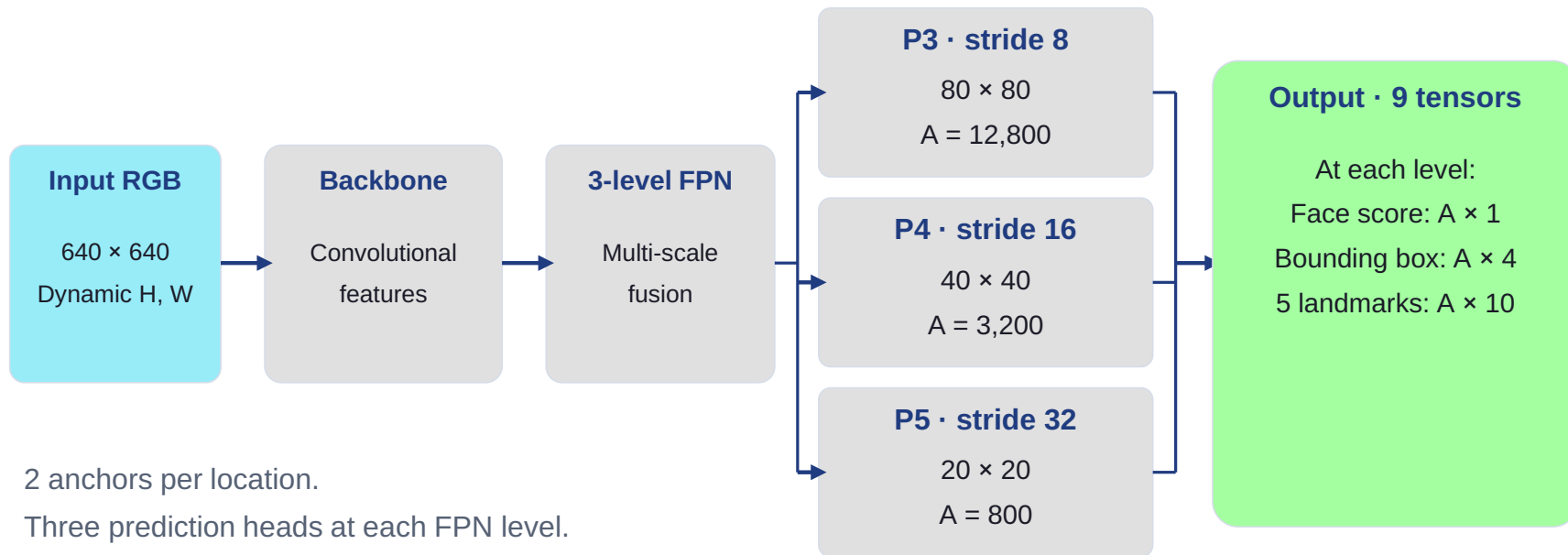
Detector stays frozen. Only selected ArcFace recognition layers are adapted.



The following diagrams reflect the exported model graphs used in the report.

RetinaFace: Actual Architecture

Frozen det_10g.onnx · typical detector input $1 \times 3 \times 640 \times 640$ RGB.



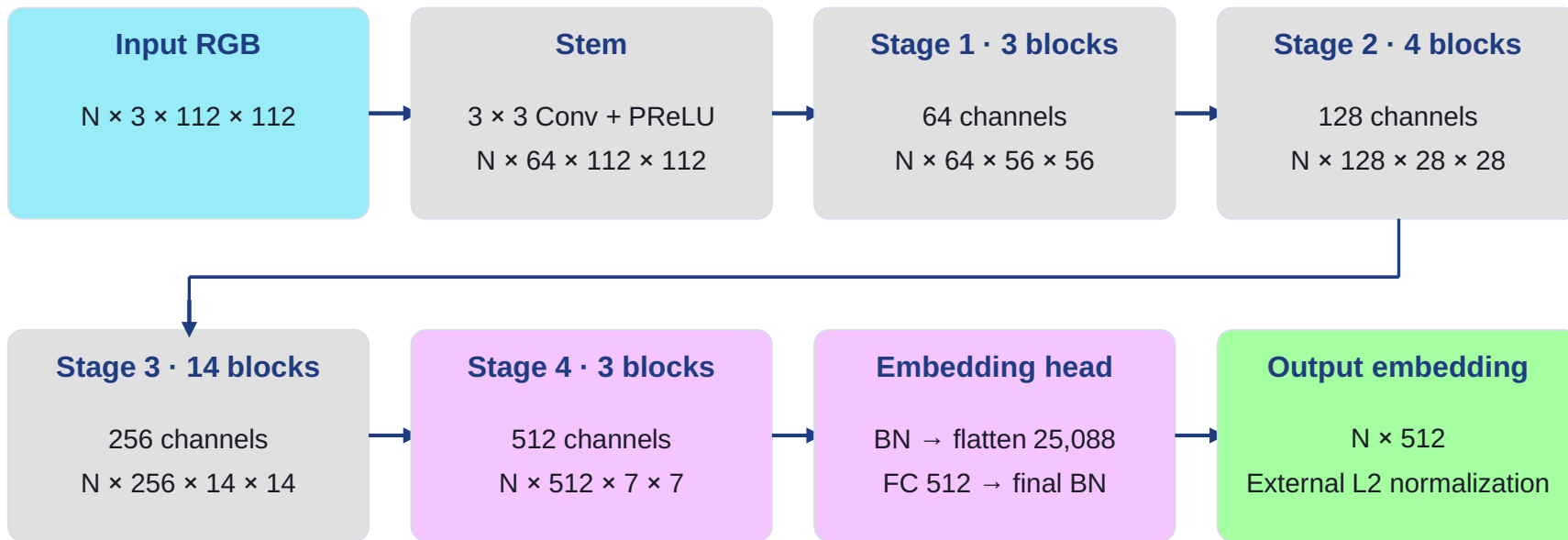
2 anchors per location.

Three prediction heads at each FPN level.

Outputs → box decoding / filtering → 5-landmark alignment → 112 × 112 RGB crop.

ArcFace: Actual iResNet-50 Architecture

w600k_r50.onnx · 24 residual blocks; block internals intentionally omitted.



E1: final block of stage 4 (layer4.2). E2: that block + embedding head. Earlier layers stay frozen.

E0 / E1 / E2: What Changes?

The reference vectors and evaluation rule are identical; the trainable scope changes.

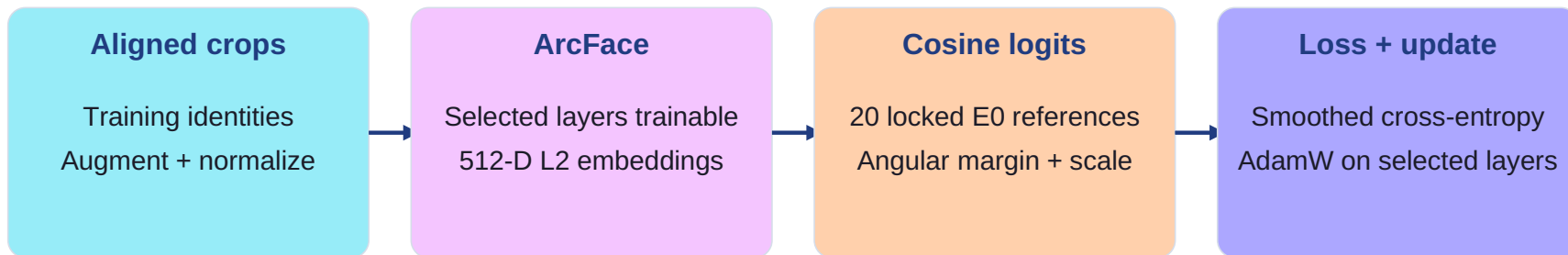
Experiment	Trainable layers	Trainable parameters
E0 · pretrained	None	0
E1 · block only	Final stage-4 residual block (layer4.2)	4,720,640
E2 · block + head	Same block + BN / FC / final BN	17,568,256

Important distinction

The 20-vector training classifier is a fixed buffer—not a learned classification head.

Training Pipeline

20 training identities per fold; gradients affect only the selected E1 or E2 layers.



Validation

All 30 gallery vectors; select by validation accuracy, then confidence gap.

Baseline is eligible

Epoch 0 is a candidate. E2 keeps E0 on folds 3, 4 and 5.

Training Recipe

Same optimization schedule and data protocol for both fine-tuned configurations.

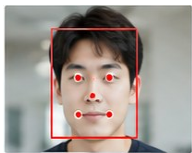
Setting	Configuration
Optimizer	AdamW · learning rate 3×10^{-5} · weight decay 10^{-4}
Schedule	Cosine decay (Tmax = 20); maximum 20 epochs; patience 5
Batch / seed	Train 32 · evaluation 64 · random seed 42
Classification loss	Scale 32; label smoothing 0.1; fixed benchmark classifier
Angular margin	0 for epochs 1–2; ramp to 0.20 over epochs 3–6
Batch normalization	Running statistics stay frozen; selected affine terms may train

Augmentations: flip, down-up blur, Gaussian blur, noise, JPEG compression, brightness / contrast.

Inference & Decision Rule

Identical identification pipeline and acceptance threshold for all three configurations.

01 Face Detection



RetinaFace returns a face box and five landmarks.

02 Alignment & Crop



112 × 112

Five landmarks align the face to a 112 × 112 RGB crop.

03 Feature Extraction

ArcFace

→ 512-D

E0 or fold-specific E1 / E2 ArcFace produces 512 features.

04 L2 Normalization

$$z / \|z\|_2$$

Normalize the embedding to unit length. No rolling buffer.

05 Gallery Matching

512-D

×

30 vectors

Compare with all 30 locked pretrained gallery vectors.

06 Decision Rule

IDENTITY

UNKNOWN

Choose the top identity if cosine ≥ 0.30 ; else UNKNOWN.

Correct = right identity AND cosine ≥ 0.30 . All queries are enrolled people; UNKNOWN means rejection.

Main Results: Pooled Held-Out Queries

5,818 queries across all six test folds; the report's saved aggregate results.

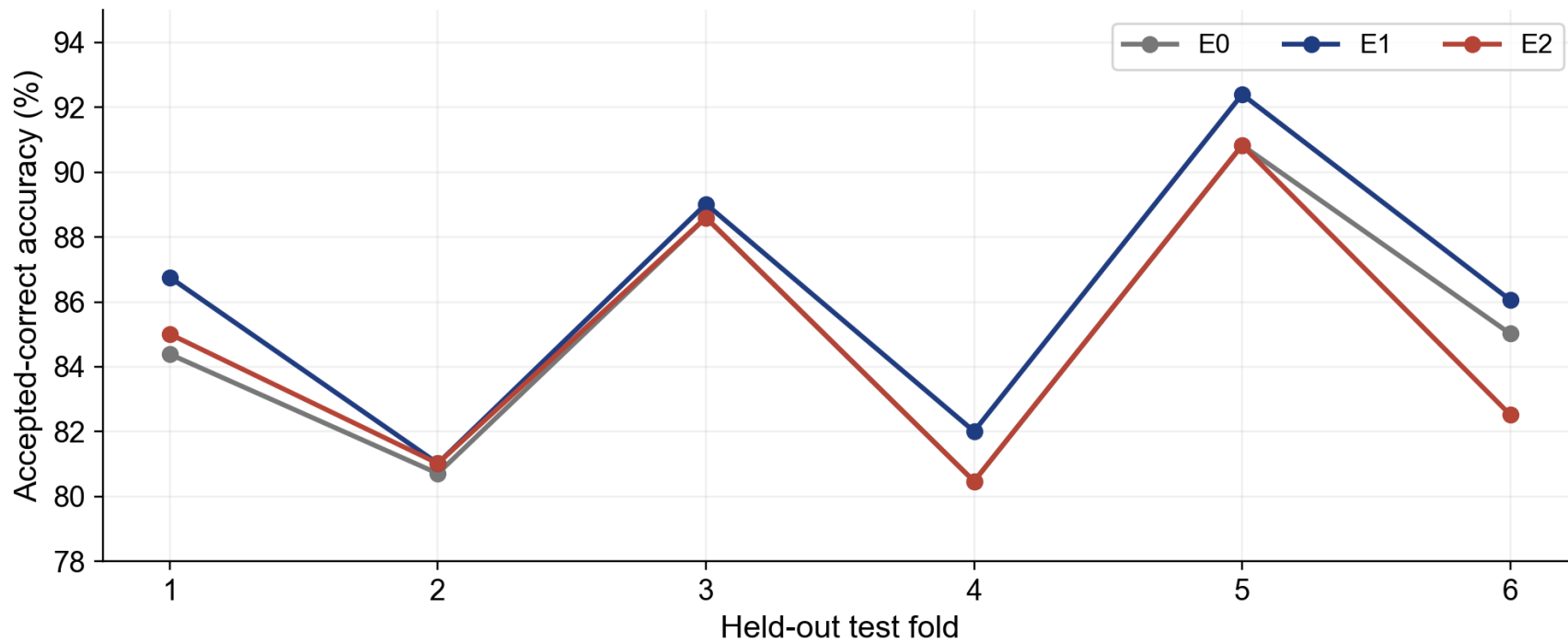
Model	Accuracy	Macro precision	Macro recall	Macro F1
E0	84.98%	100.00%	85.12%	91.67%
E1	86.18%	99.72%	86.33%	92.15%
E2	84.72%	96.43%	84.87%	89.85%

Best configuration: E1

+1.20 percentage points over E0, with 4.72M trainable parameters. E2 is 0.26 points below E0.

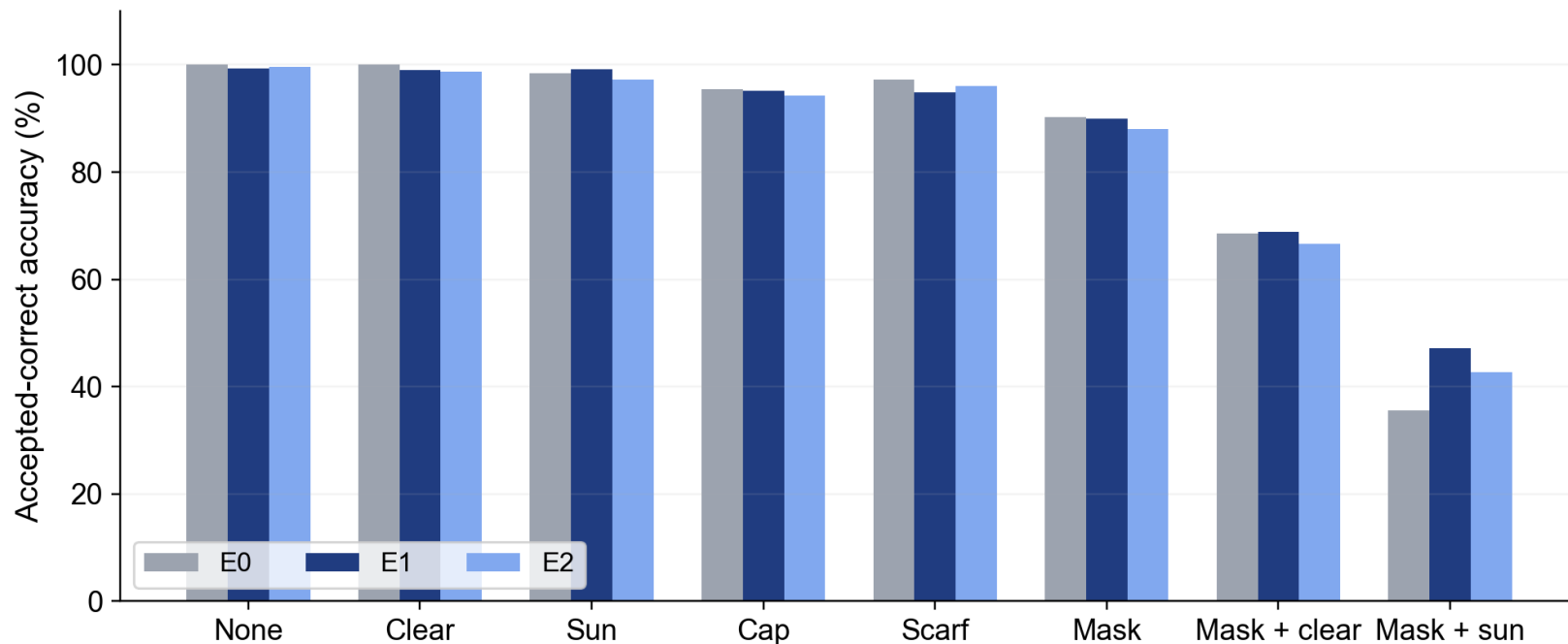
Consistency Across Six Test Folds

E1 improves over E0 on every fold; E2 improves on only two test folds.



Where E1 Helps Most

The largest benefit is severe compound occlusion—not uniform improvement in every condition.



Mask + sunglasses: 35.6% → 47.2% (+11.6 points). Scarf, mask, and clean-face results do not all improve.

Accuracy Alone Hides the Trade-Off

Lower rejection can come with more accepted wrong identities.

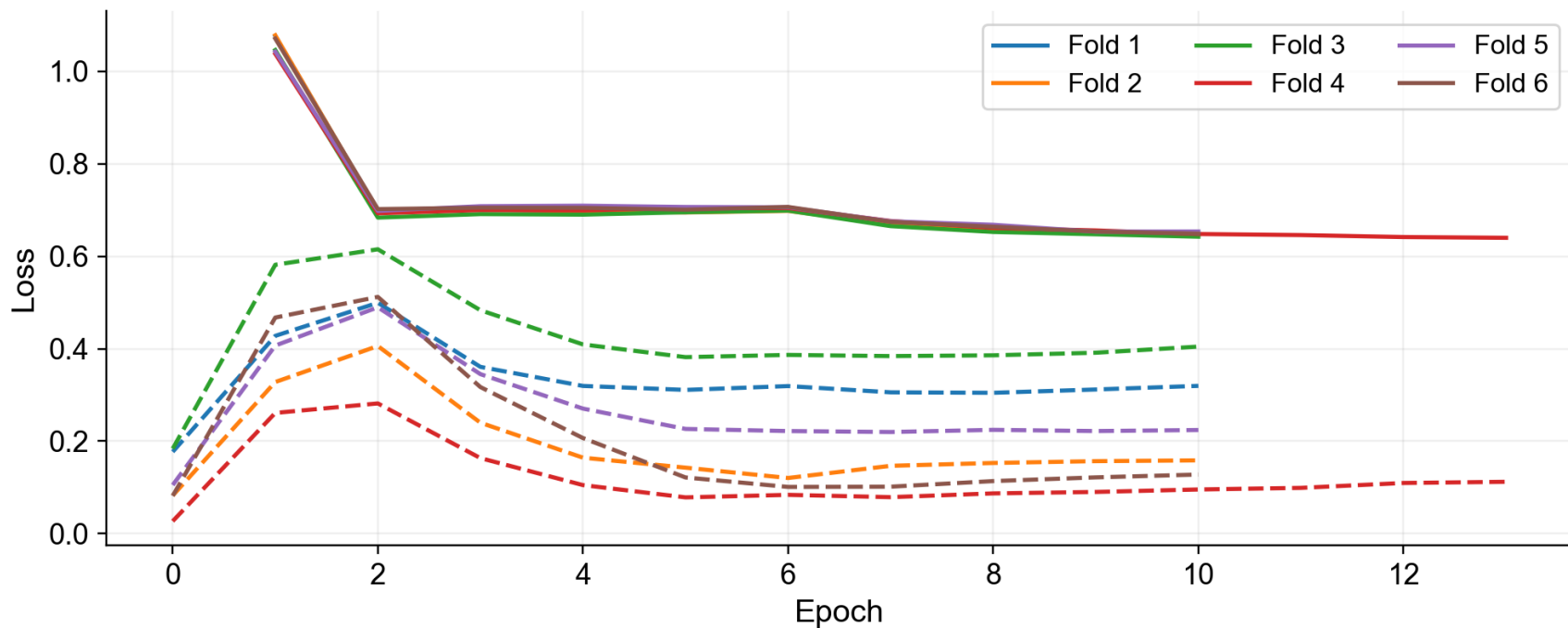
Outcome (count)	E0	E1	E2
Accepted, correct	4,944	5,014	4,929
Rejected, correct argmax	727	493	421
Rejected, wrong argmax	147	298	275
Accepted, wrong	0	13	193

Interpretation

E2 accepts 193 wrong identities (3.32%), versus 13 (0.22%) for E1.

E1: Training & Validation Loss

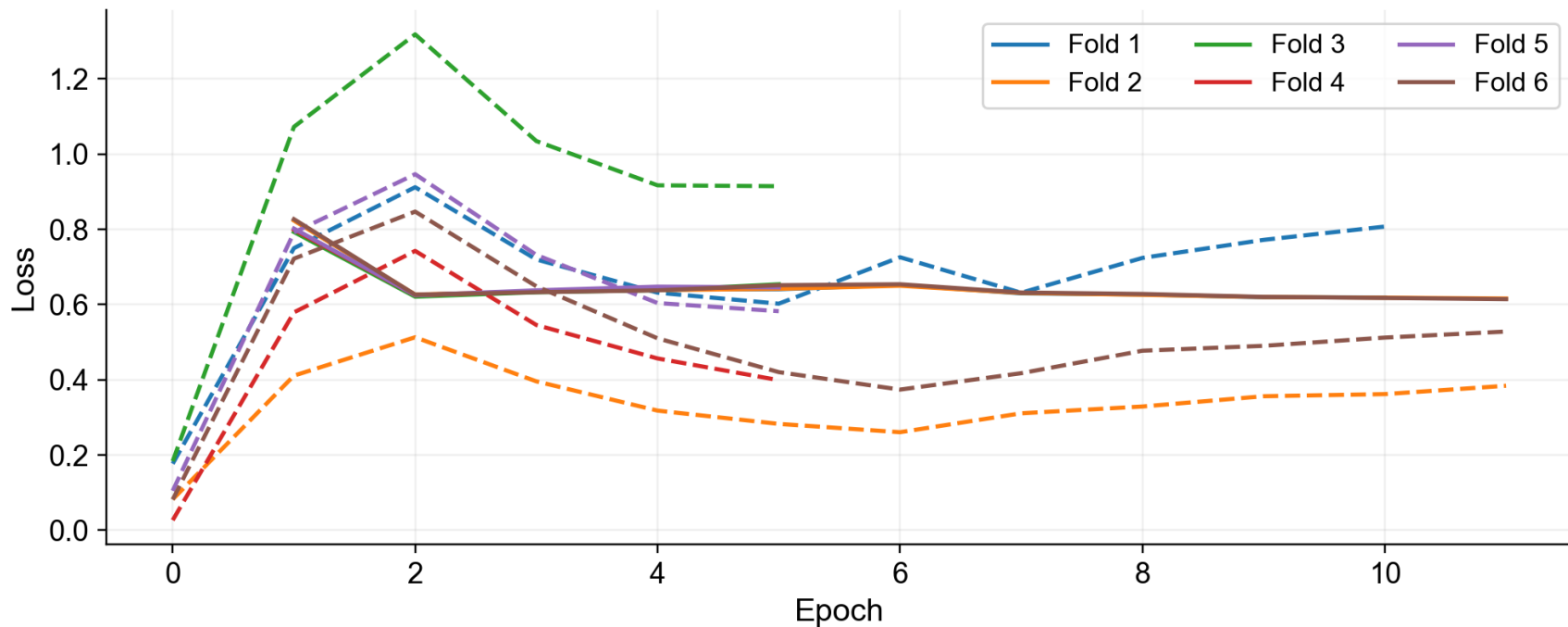
Same axes: solid = training loss; dashed = validation loss; one colour per fold.



Loss scales differ: training uses 20-class margin + smoothing; validation uses 30-class plain logits.

E2: Training & Validation Loss

Same axes: solid = training loss; dashed = validation loss; one colour per fold.



Loss scales differ: training uses 20-class margin + smoothing; validation uses 30-class plain logits.

Why More Trainable Layers Did Not Win

The block-only update is the strongest transfer configuration in this experiment.

Evidence

E1: 86.18%, improvements in all six folds.

E2: 84.72%, 193 accepted wrong labels.

Validation retained epoch 0 in three E2 folds.

The embedding FC matrix alone has
~12.85M weights.

Interpretation—not proof

With only 20 training people, the larger update may adapt too strongly to them.

A fixed pretrained gallery may amplify embedding-space drift.

These explanations need dedicated ablations; the measured conclusion is narrower.

Fine-tuning scope matters: increasing capacity does not guarantee better held-out identity transfer.

Related Work & Positioning

Complementary precedents; their reported accuracies use different datasets and protocols.

Work	Contribution	Relation to our project
Salari & Rostami, 2016	Pgu-Face partially covered-face dataset	Collection precedent; broaden covering types
Chong et al., 2022	CNN features + SVM for masked faces	Different recognition pipeline and benchmark
Montero et al., 2022	Multi-task ArcFace for masked faces	Our study varies layer scope, not task losses
Deng et al., 2019 / 2020	ArcFace / RetinaFace	Embedding and detection foundations
Our contribution: a documented local collection and a fixed-gallery, person-disjoint adaptation study.		

Limitations & Responsible Use

The results describe this collection and protocol, not unrestricted real-world recognition.

Data limitations

Only 30 people; indoor, single-sitting capture.

Scarf coverage and some conditions are incomplete.

Same-identity duplicates remain.

Previously added digital coverings are not flagged individually.

Faces remain personal data despite pseudonyms.

Evaluation limitations

All test queries are enrolled identities; no unseen-stranger rejection benchmark.

Threshold 0.30 is fixed, not a deployment guarantee.

Gallery is locked to pretrained embeddings.

Consent is academic / non-commercial and withdrawable; access is not a public license.

Key Findings

A modest, targeted update gives the best measured transfer to held-out people.

86.18%

E1 pooled accuracy

+1.20 pp

E1 improvement over E0

+11.6 pp

Mask + sunglasses gain

The new dataset targets richer occlusion variation, not greater identity diversity.

Train the final ArcFace block; keep the gallery and early representation stable.

Next: independent sessions, more participants, and unseen-identity rejection tests.

References & Reproduction

Report-aligned sources and artifacts; original deck retained as the visual template.

[Salari & Rostami · Pgu-Face · Data in Brief 9 \(2016\), 288–291](#)

[Chong et al. · Masked Face Recognition Using SVM and CNN · ICoICT 2022](#)

[Montero et al. · Boosting Masked Face Recognition with Multi-Task ArcFace · 2022](#)

[Deng et al. · ArcFace · CVPR 2019](#)

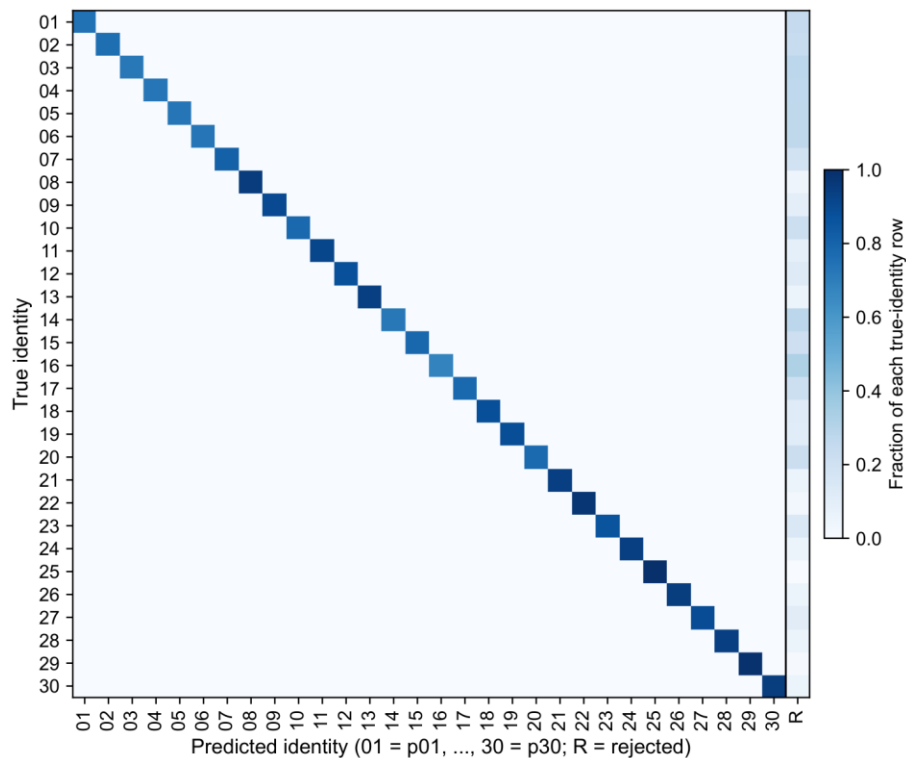
[Deng et al. · RetinaFace · CVPR 2020](#)

Project artifacts: final_technical_report.md, final.pdf, metadata/e1e2_person_folds.json, runs/e1e2/. Results reproduced from saved aggregates and histories.

[**Open our dataset folder \(Google Drive\)**](#)

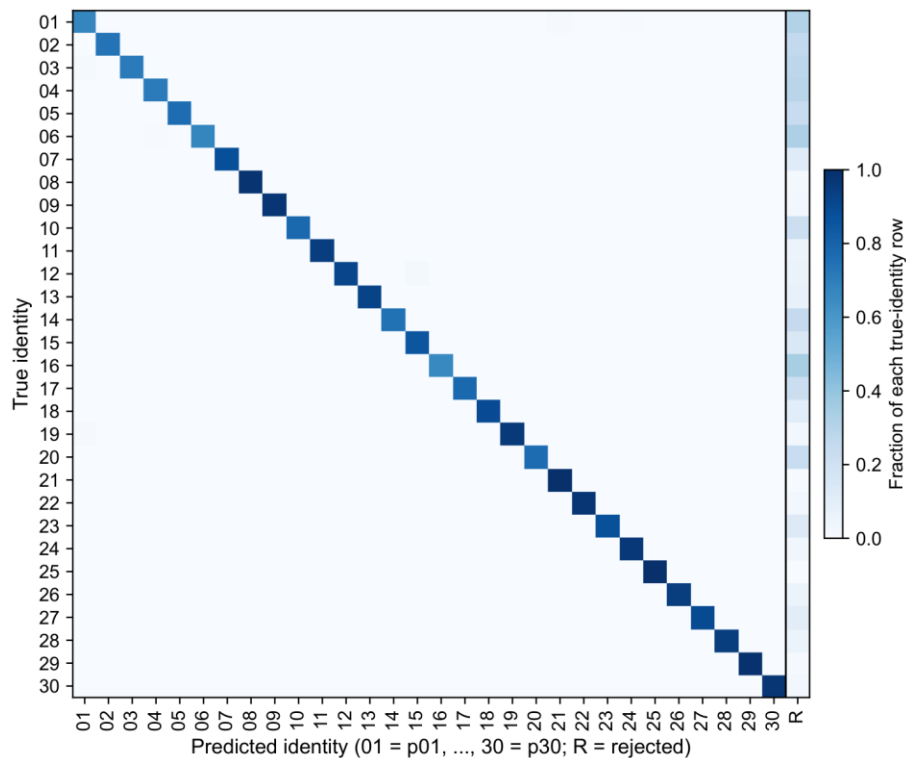
Appendix · Eo Confusion Matrix

30 identity columns + R for rejection; held-out queries pooled across six folds.



Appendix · E1 Confusion Matrix

30 identity columns + R for rejection; held-out queries pooled across six folds.



Appendix · E2 Confusion Matrix

30 identity columns + R for rejection; held-out queries pooled across six folds.

